DYNATUNE-XL SIMULATION TOOL SUITE

10 “EASY” STEPS TO HAPPY DAMPER CURVES

A Comprehensive Spring & Damper Tuning Guide

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The following 10 Step Recipe is describing our standard Approach & Recommendation for creating a **DYNATUNE-XL** Ride Model with subsequent Tuning & Optimizing or Front & Rear Axle Spring/Damper Systems. Although this document is focused on supporting the **DYNATUNE-XL SUSPENSION TUNING MODULE** user community, it does contain a range of valid generic tuning recommendations.

If interested one can find the **DYNATUNE-XL STM DEMO** Version here:
http://www.dynatune-xl.com/download-demo-stm.html

SPRING DAMPER TUNING GUIDE – 10 “EASY” STEPS

- **STEP 1: DEFINE INITIAL TARGET RIDE FREQUENCY**
- **STEP 2: OPTIMIZE BOUNCE & PITCH CENTER LOCATION**
- **STEP 3: DEFINE MAXIMUM BODY & WHEEL DAMPING AT LOW DAMPER SPEEDS**
- **STEP 4: DEFINE BODY & WHEEL DAMPING AT HIGH DAMPER SPEEDS**
- **STEP 5: TUNING OF KNEE POINT / BLOW OFF POINT**
- **STEP 6: DEFINE/VERIFY INITIAL DAMPER FORCE VELOCITY CURVE**
- **STEP 7: TUNE OVERALL BODY & WHEEL DAMPING IN FREQUENCY DOMAIN**
- **STEP 8: TUNE OVERALL BODY & DAMPING IN TIME DOMAIN**
- **STEP 9: OPTIMIZE TIME & FREQUENCY DOMAIN DAMPING CHARACTERISTICS**
- **STEP 10: EXECUTE AUTOMATIC DAMPER CREATION & OPTIMIZATION**

This document will be referring frequently to the **DYNATUNE-XL FAQ** webpage for customers:
http://www.dynatune-xl.com/support-rh.html

STEP 1: DEFINE INITIAL TARGET RIDE FREQUENCY

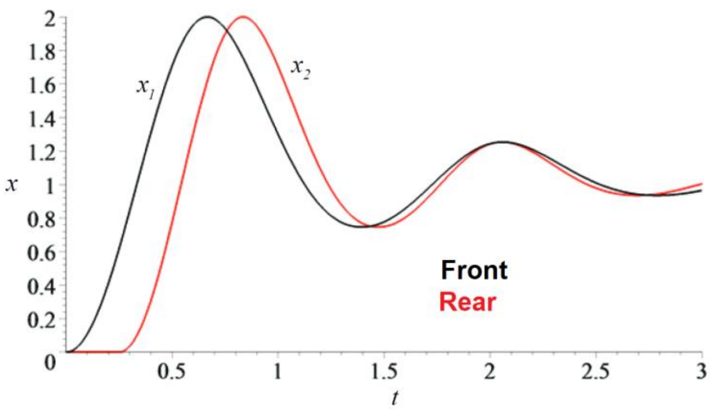
- The first essential step is to define the targeted Ride Frequency of the Vehicle. The desired Ride Frequency will ultimately define Front and Rear Axle Spring Rates and depending on the target application of the vehicle also influence the amount of damping required.

1) The “Heave” Frequency is defined by Vehicle Mass, Rate of Springs and Tires. Front & Rear Body Natural Frequencies can vary from typically around 1 Hz on a standard road car to up to well above 5 Hz on a light and heavily sprung race car. Please do look check **DYNATUNE-XL FAQ** webpage for more detailed info & tuning recommendations.

   Historically, the following general indications can be taken as a good starting point:

   - Road Car 1.1 - 1.3 Hz
   - Sporty Road Car 1.3 - 1.5 Hz
   - Sports Car 1.5 - 1.7 Hz
   - Super Car 1.7 - 2.2 Hz
   - Race Car 2.2 - 2.8 Hz
   - Formula Car > 3.0 Hz

2) Typically, the reference point is the Centre of Gravity of the sprung body mass. However, it is very common to define a Front Axle & Rear Axle Ride Frequency in order to consider and optimize specific characteristics. As a rule of thumb though one can say that typically the Rear Ride Frequency should be between 10% to 15% or around 0.1Hz higher than Front Ride Frequency as it will allow the Rear Axle Oscillation to “catch up” with the Front Axle Oscillation when driving over obstacles and thus support the subjective sensation of “flat ride” (without significant sensations of pitch angle motions).

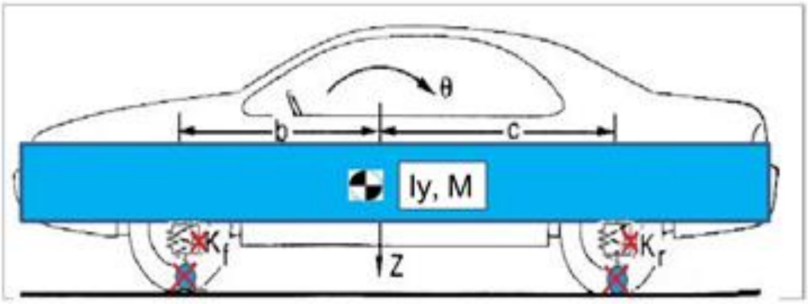


3) Assuming that the Tire Rates are given, the first selection of the Front & Rear Ride Frequencies will define the Front and Rear Wheel Rates of the Suspension. As on most vehicles the vertical stiffness of the tire (in general typically above 200 N/mm) is significantly higher than the suspension wheel rate, the key tuning parameter will be primarily front suspension and rear suspension wheel rate to achieve the desired natural body frequencies.

   Note: This document does assume that all damper related specific terms and basic physical formulations of oscillating systems with terminology like natural frequency and/or percentage critical damping (damping ratio) are fully understood. Whilst it should not be necessary to fully understand the equations of typical 2-DOF Ride Models an elementary understanding of principles of ride simulation is beneficial.

STEP 2: OPTIMIZE BOUNCE & PITCH CENTER LOCATION

- Elementary vehicle motions are Heave, Roll and Pitch. If we do eliminate for simplicity the Roll motion and do assume that the vehicle we are analysing is a single track “bicycle” model, the 2 remaining dominant vehicle movements in “vertical” direction are Heave & Pitch.



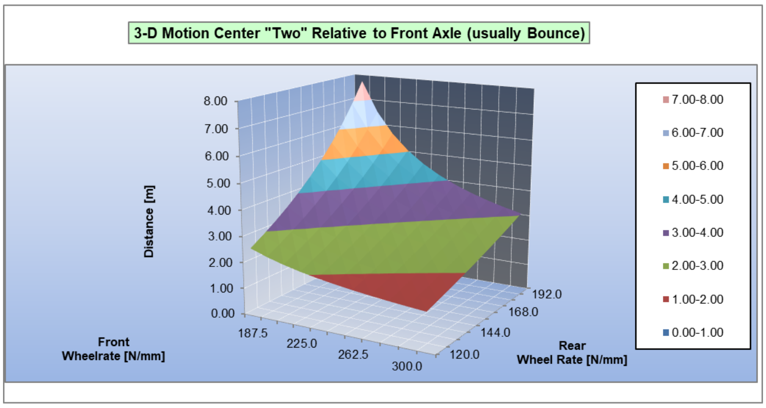
2 DOF Ride Model for a Vehicle.

1) A translational movement is equal to a rotational movement around a point with an infinite radius. Using this mechanical law does allow us to understand that for all movements of the above shown model an instantaneous centre of movement must exist. These 2 centres are commonly named Bounce & Pitch Centres. The Pitch Centre is always located within the wheelbase of the vehicle whereas the Bounce Centre is always located outside of the wheelbase of the vehicle.

   Those who are interested in all of the details should visit the **DYNATUNE-XL FAQ** webpage or click on the picture below and read the “BOUNCE & PITCH CENTRE” document for better understanding on the topic and tuning guidance.

2) Bounce & Pitch Centres do primarily influence the subjective ride perception but do also - especially on heavily sprung & light racing cars - affect the tire grip via altered aerodynamic pitch angle sensitivity.

3) The Wheel Rate (and thus in effect the rate of the coil spring in the suspension) is the principal influence factor on the location of the Bounce & pitch Centres. A careful optimization analysis between Front and Rear Axle Wheel rates will allow to optimize the location of both Motion centres.

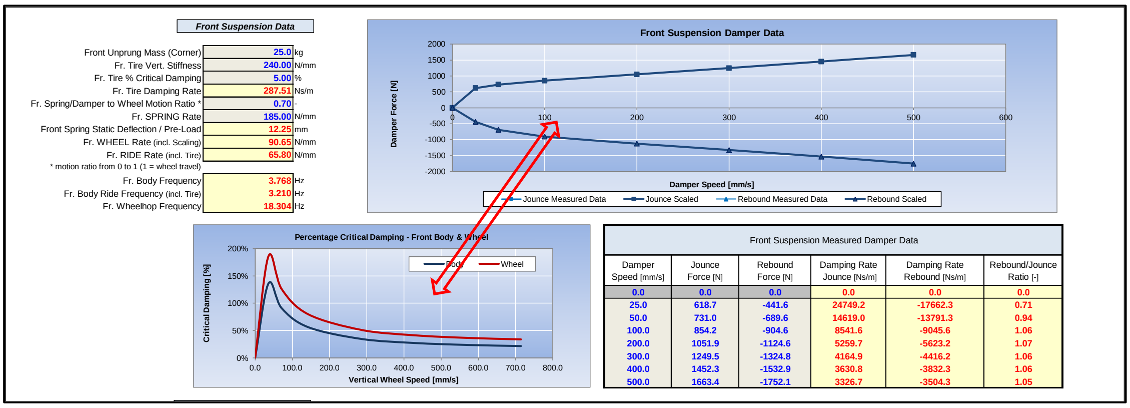


Click on picture to follow link to documents

4) As a generic tuning recommendation one can say that the Bounce Centre should be as <u>FAR</u> away as possible from the Front Axle of the vehicle and the Pitch Centre should be as <u>CLOSE</u> as possible to the Front Axle of the vehicle.

STEP 3: DEFINE MAXIMUM BODY & WHEEL DAMPING AT LOW DAMPER SPEEDS

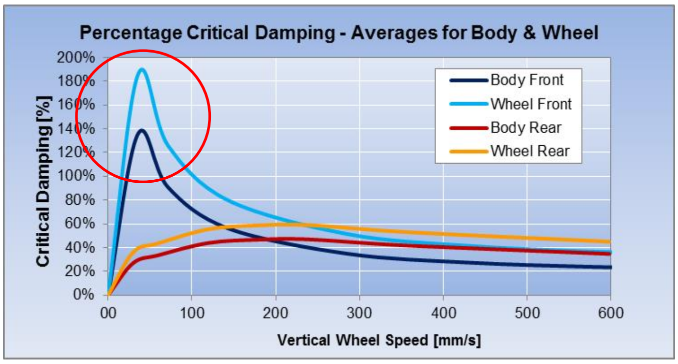
- In the next step, a first target for the Damper Force-Velocity Curves will have to defined and eventually optimized. Unfortunately, since we are starting the process from a vehicle perspective we cannot define directly the actual damper curve but we must create/define a target curve for the Damping Ratio (% Critical Damping) as a function of the vertical wheel velocity.



- The first step in the damper optimization exercise is to set targets for the Damping Ratio – which is also known as Percentage Critical Damping – for the sprung Body. In the field of Damper Tuning, one should know and understand 2 particular values:

    - 100%. A Damping-Ratio of 100% does mean that the vehicle is critically damped, which does mean that it will not perform an oscillation after a vertical input but will move (slowly) to its final position without any overshoot.

    - 70%. A Damping Ratio of 70% will allow the vehicle to do 1 positive overshoot and 1 negative undershoot (half-) oscillation relative to its final position after a vertical input.



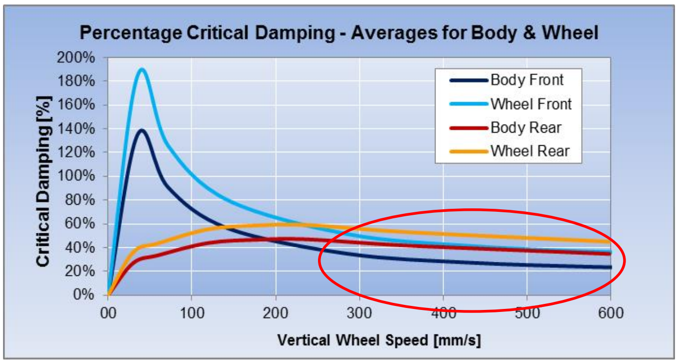
    - Typically, the maximum value for Damping Ratio of the Body should occur at a vertical suspension velocity in the range of 100mm/s to 200mm/s

- Some general recommendations for Maximal Overall Damping Ratios for the Body:

    - Road Car 30% - 40%
    - Sporty Road Car 40% - 60%
    - Sports Car 50% - 70%
    - Super Car 60% - 80%
    - Race Car 70% - 100%
    - Formula Car > 80%

STEP 4: DEFINE BODY & WHEEL DAMPING AT HIGH DAMPER SPEEDS

- Next to defining targets for the Maximum Damping Ratio values for the Sprung Mass (Body) it is equally important to set a target for the Wheel Damping Ratio at high vertical wheel velocities. For passive Damper Systems it is not easily possible tuning Wheel and Body Damping Ratio independently at the occurring damper speeds.



Whereas indicative damping values for the sprung mass are quite commonly known & categorized, the generic guidance for unsprung mass target damping values at higher vertical suspension velocities (> 500mm/s) are less clearly defined.

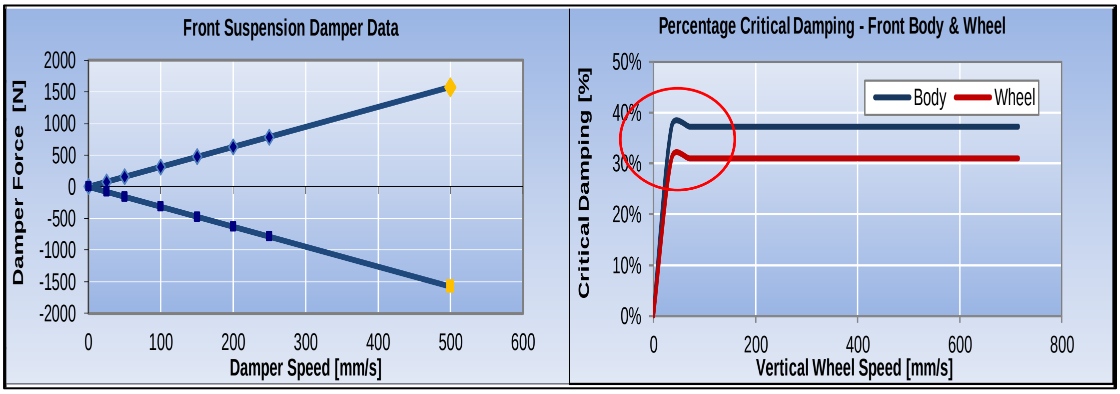
Typically, one can say:

- Road Car — Range: Jounce 10% to Rebound 40%
- Race Car — Range: Jounce 20% to Rebound 100%

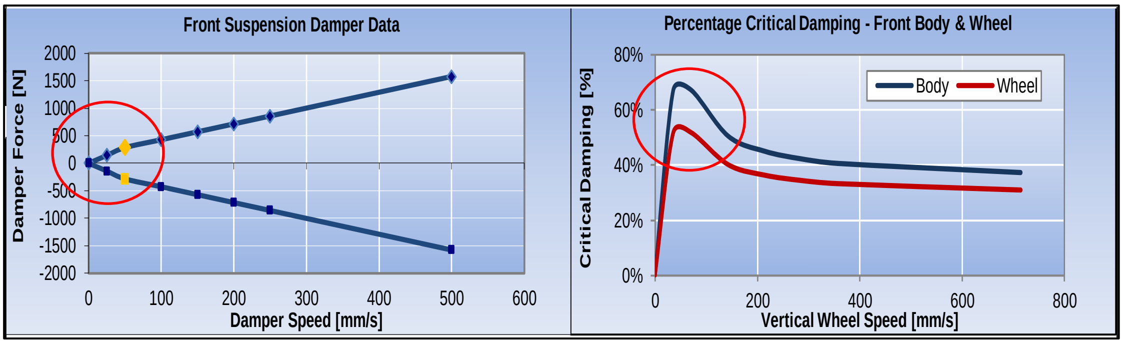
STEP 5: TUNING OF KNEE POINT / BLOW OFF POINT

- The often-conflicting requirements for sufficient damping of the body at low damper speeds and typical body natural frequencies does especially on vehicles with low mass and high spring rates require a modification to the damper system in which in essence above a certain damper velocity the amount of damper force is being limited by an internal blow off valve. This will change the damper force curve from linear to non-linear

<u>Linear Damper:</u>



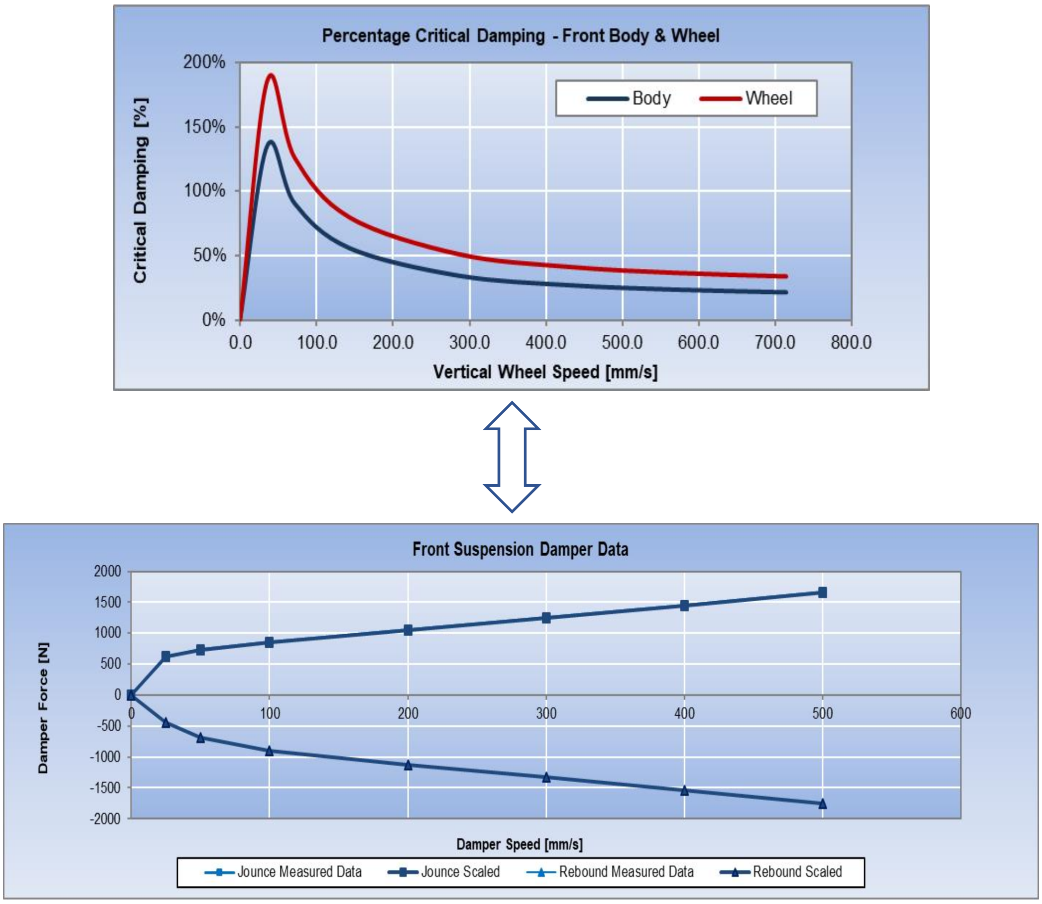
Non-Linear Damper:



Tuning of the Knee-Point (or Blow-Of) does typically take place in a vertical wheel speed velocity range of 50mm/s to 150mm/s and is a fundamental instrument to allow a strong body control at low frequencies whilst permitting a compliant control at higher frequencies (for instance for allowing smooth running over obstacles).

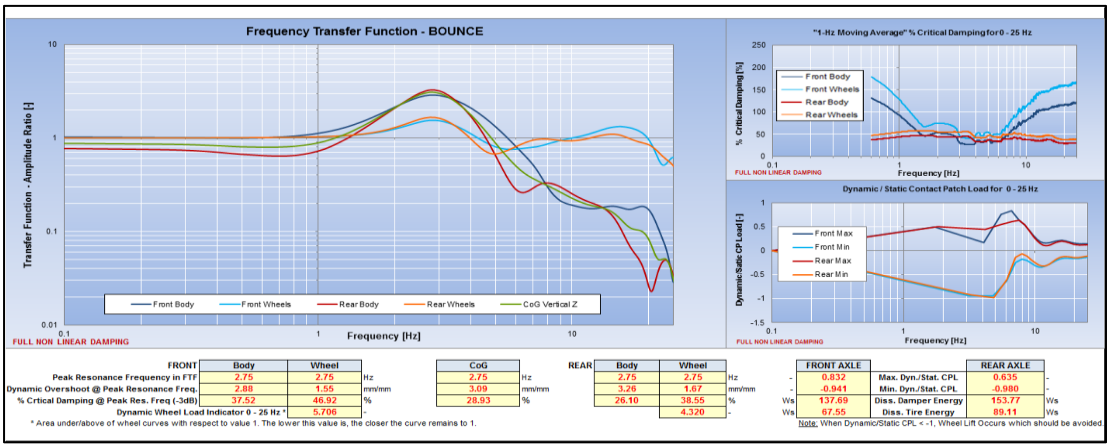
STEP 6: DEFINE/VERIFY INITIAL DAMPER FORCE VELOCITY CURVE

- Applying in all of the steps above the correct numbers and considering the motion ratio of the Damper to the Wheel will allow to reverse engineer a Damper Force – Damper Velocity Curve which can be used as a starting point for the following steps of refinement. The user of the **DYNATUNE-XL STM MODULE** can modify the damper input data in such a manner that the required Wheel Damping Ratio Curve can be matched.



STEP 7: TUNE OVERALL BODY & WHEEL DAMPING IN FREQUENCY DOMAIN

- Whereas Step 1 to Step 6 do represent the usual steps of “classic” Damper Force – Damper Velocity Curve analysis, we must now start to set targets for the spring/damper system in the typical operating Frequency Range from 0 Hz to 25 Hz. Without entering into complex details, it is important to understand that Driving Velocity and Obstacle Amplitude define the Excitation Frequency of the Spring/Damper/Axle System. Hence a Frequency Domain Analysis has to be executed, which does in essence tell you how much response there is to a given input at a defined frequency. Also known as Frequency Transfer Function & 4 Poster Test. The test can be executed in a “Bounce” or “Pitch” Mode. Here we do concentrate on the Bounce mode.

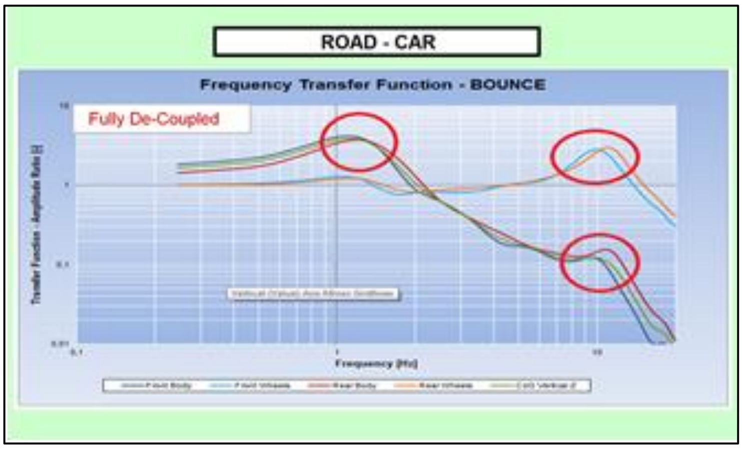


Click on picture to follow link to website

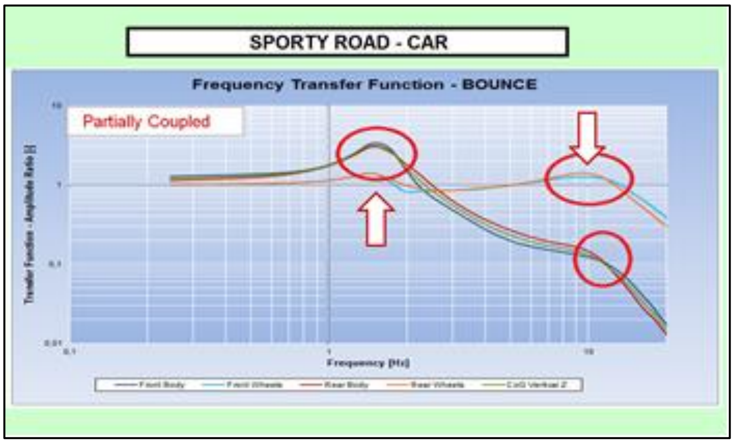
In this optimization procedure the Ride Model from Step 1 is excited by a sinusoidal road input signal with a decaying amplitude over frequency. By changing the amplitude of the sinusoidal the effective vertical wheel velocity (aka damper velocity) can be tuned to match the vehicle mission. The results indicate how much the vehicle does react at a given frequency to the input signal amplitude at that same given frequency. At this point it is highly recommended to keep at hand the extensive info provided in the **README** Section of the **STM** Module.

In the picture above (left) one can for instance see, that the front body at an exciting frequency of 3 Hz for each x1 mm of road input does react with x3 mm of vertical displacement – in fact an amplification factor of 3. At an exciting frequency of 10 Hz however, this relationship is approximately only 0.2 and thus approximately 15 times less. By changing the amplitude of the sinusoidal input signal

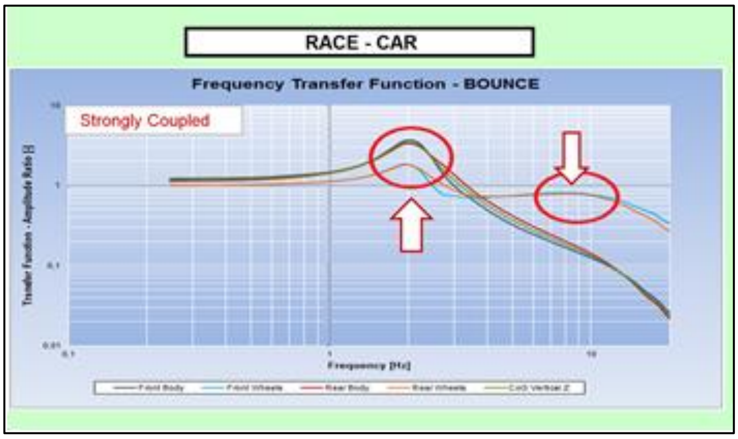
1) Types of Transfer Functions. Depending on the Wheel & Tire Rates, Vehicle Mass and Inertia’s the Frequency Transfer Functions can show more or less pronounced peaks at Wheel & Body Natural Frequencies.



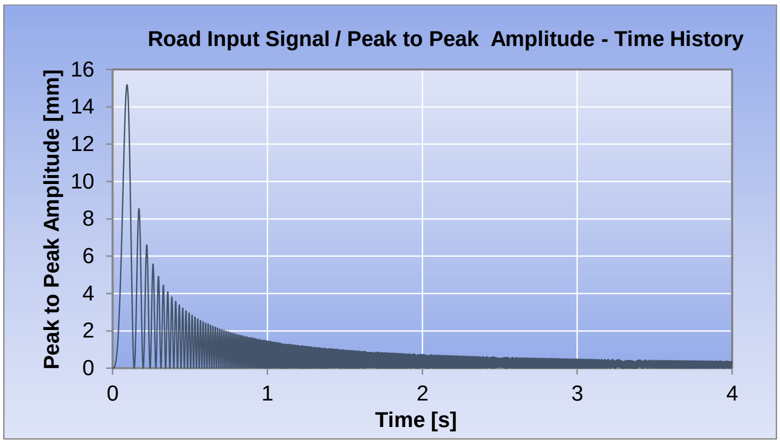
On a Sporty Road Car tendentially the Body and Wheel Hop Frequency are higher damped:



Whereas on a race car, due to the stiffer the Suspension Rates, the Wheel-Body System does become more “couples” with corresponding changes in Frequency Behaviour, mainly characterized by a “disappearing” Wheel Hop Mode and a combined Body & Wheel “Peak” reaction.



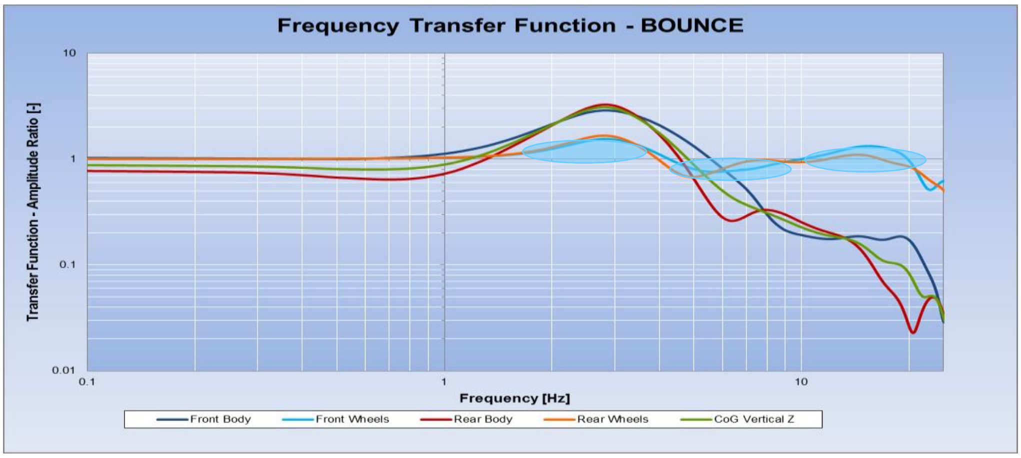
2) Road Input Profiles: Depending on the mission of the vehicle one should select an appropriate Road Signal for the Test. For those who are interested, it is highly recommended to carefully read the **README** Section of the **STM Module.** The Standard Setting of the Test Procedure has been selected to fit all sporty vehicles well and is derived from years of real world 4-Poster Testing.



3) The Frequency Transfer Function is typically used for defining the overall amount of Damping. A typical recommendation is to keep the transfer functions for Body and Wheel as close as possible to 1. As it is difficult to optimize both Body and Wheel simultaneously, one will have to decide which one is more important given the mission of the vehicle. Typically, Body Control is preferred in Comfort and Aerodynamic related vehicle setups, whereas Wheel Control is always favoured when looking for mechanical grip.

   a. <u>Optimizing for Mechanical Grip</u>:

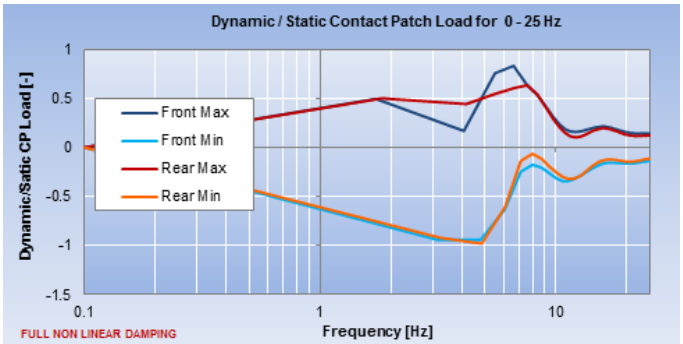
      In the Graph below one can see as example frequency ranges where the Transfer Function of the Front Wheels is diverting from 1. Optimizing Wheel behaviour will involve changing the Transfer Function.



A commonly used metric to evaluate the quality of the Transfer Function is measuring the surface of the area which is enclosed by the transfer function and the nominal starting value 1 (blue bubbles). The lower the number the better:

Dynamic Wheel Load Indicator 0 - 25 Hz *	5.706	-

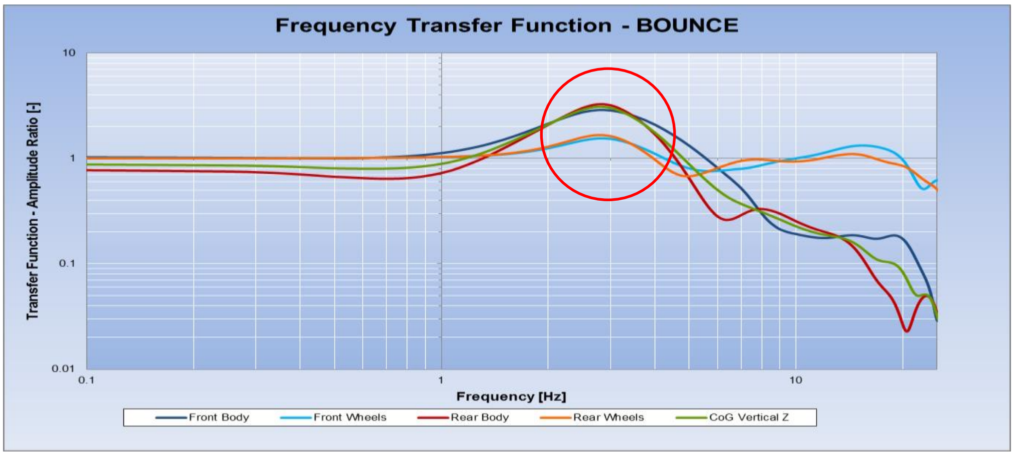
This metric above is quite well correlating with the Minimum and Maximum Dynamic Tire Loads occurring during this test. The objective would here be to minimize the dynamic changes, especially trying to minimize tire unloading.



	FRONT AXLE		REAR AXLE	
-	0.832	Max. Dyn./Stat. CPL	0.635	-
-	-0.941	Min. Dyn./Stat. CPL	-0.980	-

- <u>Optimizing for Body Control</u>

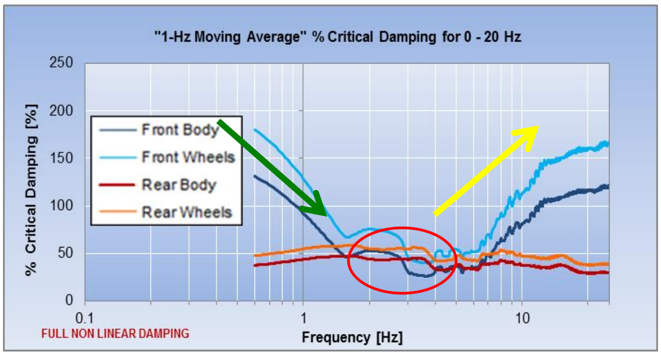
When optimizing for Body Control – typically on High Downforce Formula Cars, the objective will be to minimize Front and Rear Body Peaks in the Transfer Function.



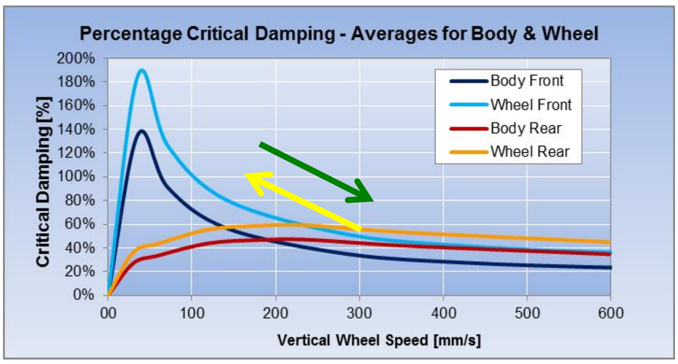
With corresponding metric values for instance for the Front Body:

FRONT	Body
Peak Resonance Frequency in FTF	2.75

When looking at the Percentage Critical Damping (Damping Ratio) in during the Test Procedure one can see that at the Peak Frequencies of Front and Rear Body the average amount of Damping is around 40%. This is a typical value and hence a good starting point for any optimization procedure.

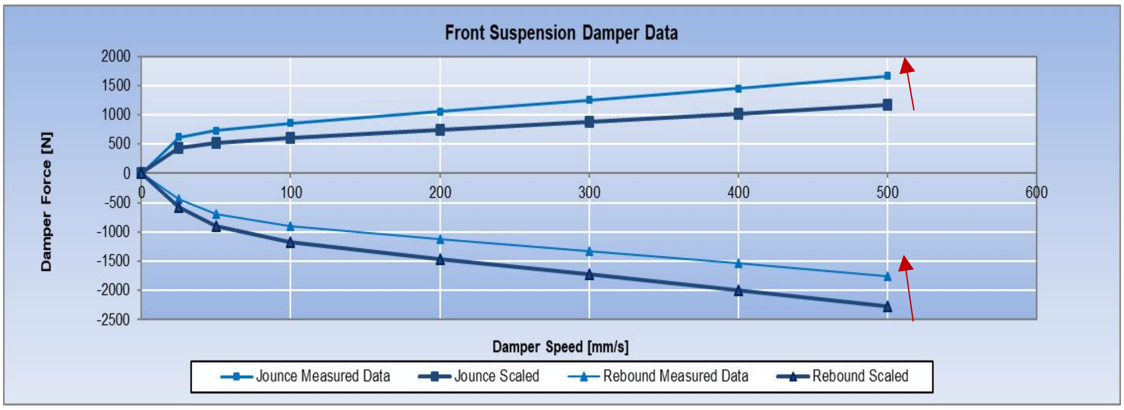


The Green and Yellow Arrows indicate how the Body Damping is changing over the Frequency and allow to visualize the corresponding "movement" in the Classic Damping Chart from Step 3 in the graph below:

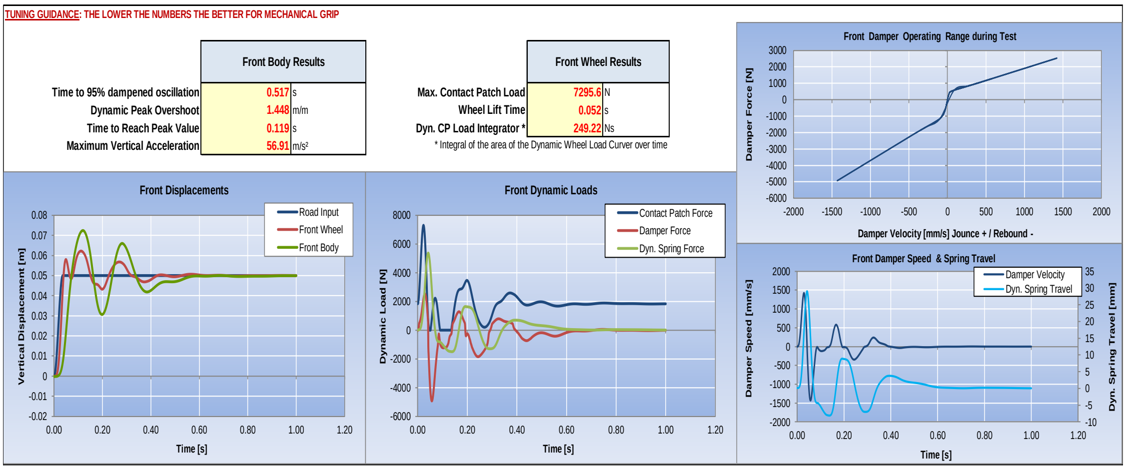


STEP 8: TUNE OVERALL BODY & DAMPING IN TIME DOMAIN

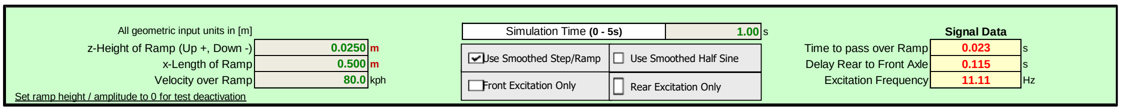
- After having established the required amount of overall damping in the 4 Poster Test, it is in this next step necessary to divide the total amount of damper force into the correct amounts of jounce and rebound damping. This exercise is typically executed in the Time Domain (Obstacle Test) and does primarily exist out of a potential re-distribution of the Jounce & Rebound Damping keeping the TOTAL amount of Damping constant.



1) Typically, any Damper Tuning in the Time Domain is being executed either with a Ramp Road Input or a Have Sine Input. Both input signals do permit to define specific performance metrics which do allow objective comparison – again for grip, body control or comfort - between different setup variants.



DYNATUNE-XL STM Module does provide a range of positive & negative ramps/havsin where analysis can be executed up to the point of loss of contact between the tire and road.

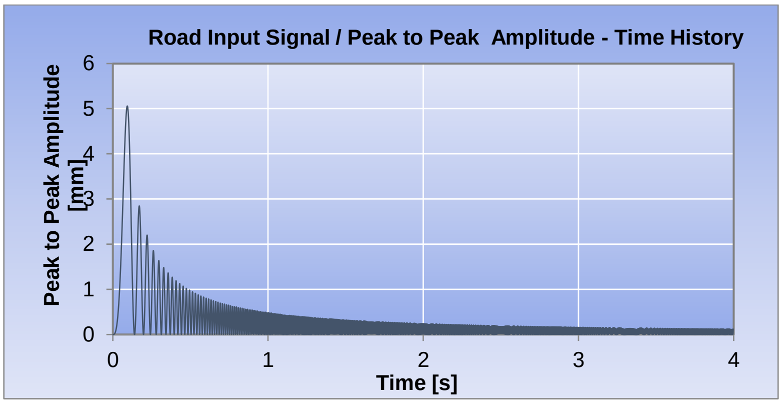


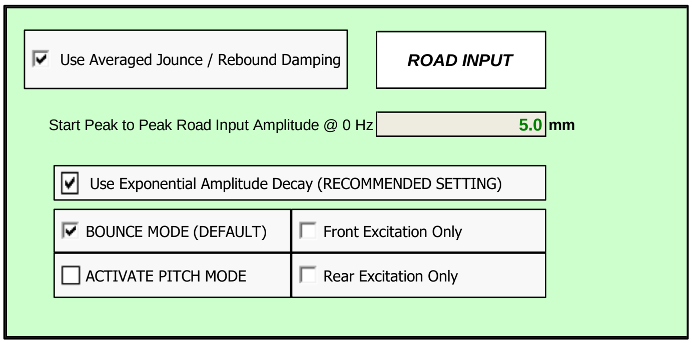
2) Damper Tuning in the Time Domain does provide other challenges than working in the Frequency Domain. Axles are exited with delays and oscillations have to be judged. On top of that changes in jounce and rebound damping do immediately affect the performance metrics. It is fair to say that this tuning step does require a lot of iterations which can be quite time consuming. Also, one must always check the results in the Frequency Domain. The complexity of these interactions will lead to the next recommended step.

STEP 9: OPTIMIZE TIME & FREQUENCY DOMAIN DAMPING CHARACTERISTICS

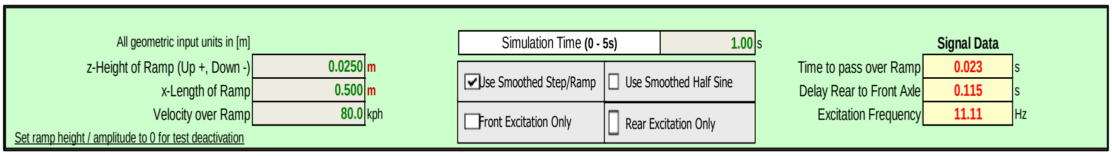
- Optimizing the Spring Damping Characteristics in both the Time Domain as the Frequency Domain can be a time-consuming exercise which – if not executed carefully and in a structured manner – can lead to inconsistent results. The following recipe is therefore recommended. It will save time and lead effectively to the desired results.

  1) Confirm / Select Road Input Profile for 4-Poster Test and Test Mode





  2) Confirm / Select Road Input Profile for Obstacle Test and Confirm / Set Vehicle Speed



  3) Select / Define Key Performance Metrics that do reflect best the target of your optimization Effort both for 4 Poster Test and Obstacle Test:

     - Mechanical Grip (Tire Dynamic Loads)
     - Body/Platform Control (Aerodynamics)
     - Comfort (Accelerations)

- Available 4 Poster Test KPI:

FRONT	Body	Wheel		CoG		REAR Body	REAR Wheel	
Peak Resonance Frequency in FTF	2.75	2.75	Hz	2.75	Hz	2.75	2.75	Hz
Dynamic Overshoot @ Peak Resonance Freq.	2.60	1.56	mm/mm	3.11	mm/mm	3.50	1.71	mm/mm
% Crtical Damping @ Peak Res. Freq (-3dB)	34.67	45.99	%	28.33	%	26.67	36.60	%
Dynamic Wheel Load Indicator 0 - 25 Hz *		4.663	-				4.718	-

* Area under/above of wheel curves with respect to value 1. The lower this value is, the closer the curve remains to 1.

	FRONT AXLE		REAR AXLE	
-	0.817	Max. Dyn./Stat. CPL	0.674	-
-	-0.986	Min. Dyn./Stat. CPL	-0.979	-
Ws	131.14	Diss. Damper Energy	156.93	Ws
Ws	69.04	Diss. Tire Energy	90.24	Ws

Note: When Dynamic/Static CPL < -1, Wheel Lift Occurs which should be avoided.

- Available Obstacle Test KPI

	Front Body Results	
Time to 95% dampened oscillation	0.517	s
Dynamic Peak Overshoot	1.448	m/m
Time to Reach Peak Value	0.119	s
Maximum Vertical Acceleration	56.91	m/s²

	Front Wheel Results	
Max. Contact Patch Load	7295.6	N
Wheel Lift Time	0.052	s
Dyn. CP Load Integrator *	249.22	Ns

* Integral of the area of the Dynamic Wheel Load Curver over time

4) In alternative to 3) **DYNATUNE-XL STM MODULE** does come with a custom developed COST Function which does combine a range of Metrics into one single Value. This Metric is called the Performance Index (PI) and is available for the Frequency Domain Tests and the Time Domain Test. The structure of the PI is in such a manner that a lower value is better:

Frequency Domain Perf. Index *	1048.76	-

* Value does depend on PI Factor settings

Time Domain Performance Index *	3699.67	-

* Value does depend on PI Factor settings

5) Define / Set Contribution of Key Contribution Metrics to Performance Index:

STM PERFORMANCE INDEX - CALIBRATION FACTORS			
Platform Control / Aerodynamic Stability	8.0	68.6	% Contribution to PI-Value
[Recommended Range 0 - 10]			
Wheel Control / Mechanical Tire Grip	10.0	26.1	% Contribution to PI-Value
[Recommended Range 0 - 10]			
Driver Comfort / Vertical Accelerations *	1.0	5.2	% Contribution to PI-Value
[Recommended Range 0 - 10]	* Time Domain Only		

6) Start Optimizing by scaling the Damper Force Velocity Curves (& if desired Spring Rates) by using the Tuning Interface:

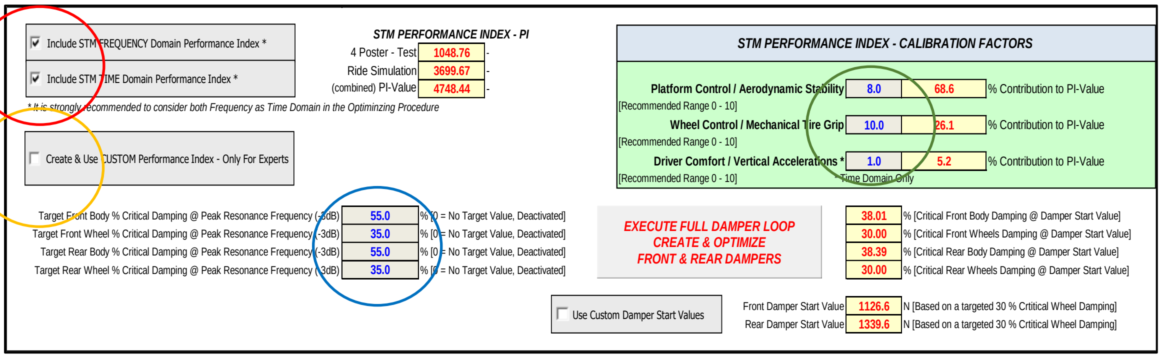
CALCULATE / F9

Fr. Jounce Damping Scaling Factor	0.70	Rr. Jounce Damping Scaling Factor	1.00
Fr. Rebound Damping Scaling Factor	1.30	Rr. Rebound Damping Scaling Factor	1.00
Fr. Spring Rate Scaling Factor	1.00	Rr. Spring Rate Scaling Factor	1.00

The Spring & Damper Scaling Factors mutiply the spring rates and the measured damper data by the selected factor - for jounce & rebound separately.

STEP 10: EXECUTE AUTOMATIC DAMPER CREATION & OPTIMIZATION

- **DYNATUNE-XL SUSPENSION TUNING MODULE** does come with a Unique Automatic Damper Curve Creation & Optimization Feature. In essence STEP 3 TO STEP 9 are executed automatically. The Feature does come with a standalone interface:



The following User Actions are necessary:

1) The User MUST define which of the 2 available PI are being considered in the exercise – Red Circle

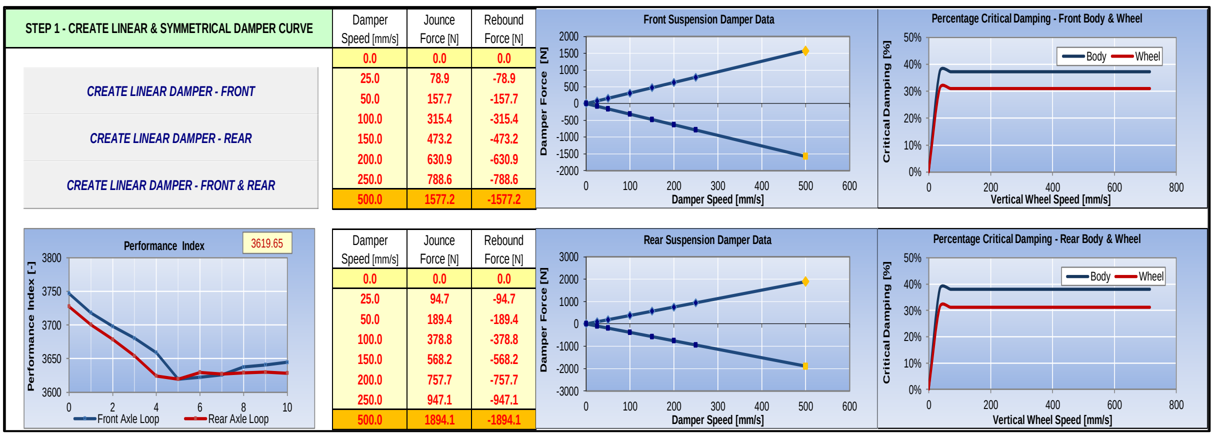
2) The User CAN define a custom Cost Function which is only recommended for Experts – Orange Circle

3) The User CAN define initial Target Values for Wheel and Body Damping Ratio's. This is not a must. If set to 0 the automatic feature will not consider any of those and optimize to the remainder of the constraints – Blue Circle

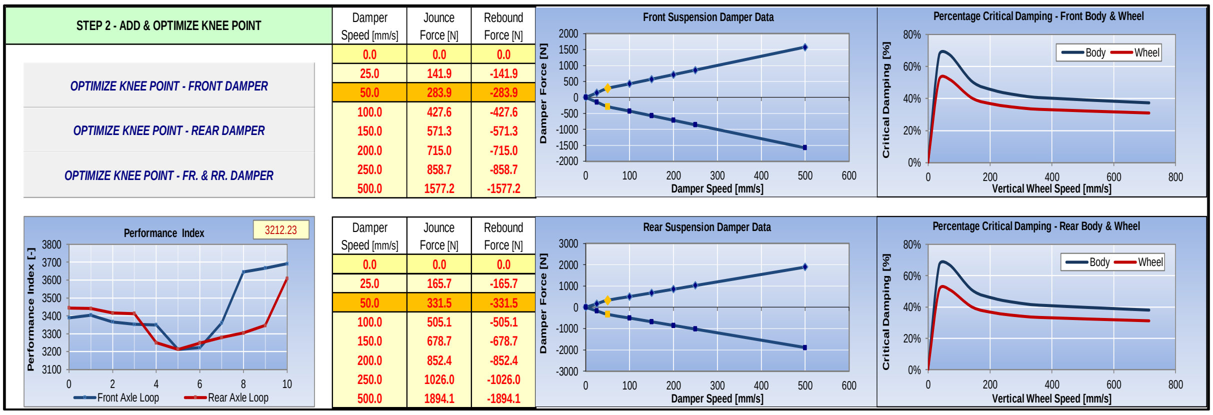
4) The User MUST define the percentage distribution of the evaluation metrics used for the optimization (Grip / Platform / Comfort).

- Once these parameters have been defined the procedure can be started.

1) In a first step the tool will create a linear & symmetric damper curve which will be meeting as good as possible the required objectives. The progress of the PI can be seen in the graph left:



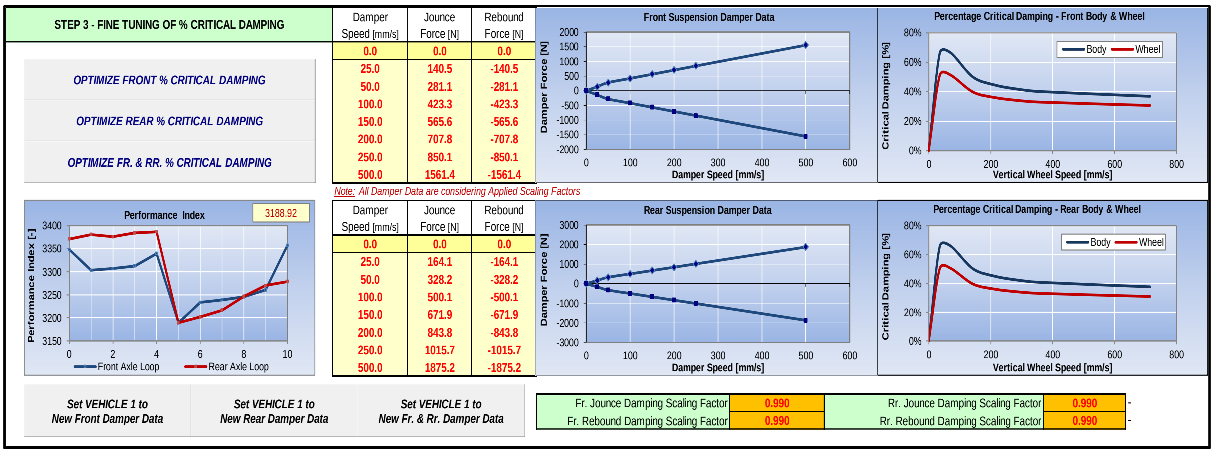
2) In a second step the tool will create – if necessary – a non-linear damper with a specific knee point at a by the user to be defined damper velocity. The damper will still be symmetric.



Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	141.9	-141.9
50.0	283.9	-283.9
100.0	427.6	-427.6
150.0	571.3	-571.3
200.0	715.0	-715.0
250.0	858.7	-858.7
500.0	1577.2	-1577.2

Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	165.7	-165.7
50.0	331.5	-331.5
100.0	505.1	-505.1
150.0	678.7	-678.7
200.0	852.4	-852.4
250.0	1026.0	-1026.0
500.0	1894.1	-1894.1

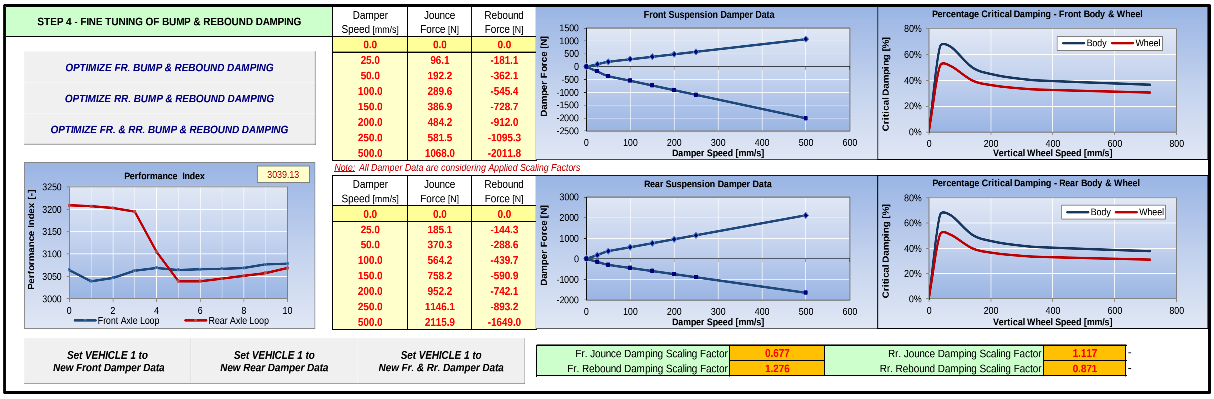
3) In the third step the tool will fine tune the results from step 1) and step 2) by adjusting scaling factors. The User can see the progress by the scaling factor and the value of the PI:



Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	140.5	-140.5
50.0	281.1	-281.1
100.0	423.3	-423.3
150.0	565.6	-565.6
200.0	707.8	-707.8
250.0	850.1	-850.1
500.0	1561.4	-1561.4

Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	164.1	-164.1
50.0	328.2	-328.2
100.0	500.1	-500.1
150.0	671.9	-671.9
200.0	843.8	-843.8
250.0	1015.7	-1015.7
500.0	1875.2	-1875.2

4) In the final fourth step the tool will concentrate on optimizing the amount of Jounce and Rebound Damping. The amount of overall damping will remain the same. Again the User can see the progress by looking at the resulting scaling factors and the progress of the PI Value:



Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	96.1	-181.1
50.0	192.2	-362.1
100.0	289.6	-545.4
150.0	386.9	-728.7
200.0	484.2	-912.0
250.0	581.5	-1095.3
500.0	1068.0	-2011.8

Damper Speed [mm/s]	Jounce Force [N]	Rebound Force [N]
0.0	0.0	0.0
25.0	185.1	-144.3
50.0	370.3	-288.6
100.0	564.2	-439.7
150.0	758.2	-590.9
200.0	952.2	-742.1
250.0	1146.1	-893.2
500.0	2115.9	-1649.0

DYNATUNE – XL

DYNATUNE-XL is the registered name of a suite of core skill **MS EXCEL ®** based Engineering and Simulation Tools.

The **DYNATUNE-XL** Tool Suite does provide Professional Engineering Tools covering the most Important Aspects of Vehicle Dynamics. All Tools aim to achieve a Maximum of Results with a Minimum of Input Data allowing quick Setup Checks or - if wanted - more complex Generic Parameter Studies. Being a fully **MS EXCEL ®** based Tool does significantly reduce the application threshold for many engineers and technicians. MS Excel is available on most computers as part of **MS OFFICE ®** and widely supported in business applications**.**

SOFTWARE REQUIREMENTS & LICENSE MANAGEMENT

Software requirements for **RELEASE 8.0** are **Full** Versions (incl. latest updates) of **MS EXCEL ® 2007, 2010, 2013, 2016** or **2019** or **Office/365** with a **MS Windows ® XP, Windows Vista, Windows 7 Starter, Windows 7, Windows 8** or **10 Operating System**.

All Modules of **DYNATUNE-XL** come as a compiled executable (*.exe) binary file which will call **MS EXCEL®** as a separate stand-alone instance. Source code is copyright protected and safe data handling is guaranteed through secure binary files.

Standard Customer Licenses are typically valid for the use of the workbooks (and ALL user-saved variants) on 1 computer and for 1 user only without a timing constraint.

The protection software does offer to the customer next to the security of encoded binary data handling also - by means of a unique License Key Verification Procedure - a state–of-the-art data protection.

License support is available for the latest releases only and as there is no annual maintenance fee existing clients with older product releases can acquire "upgrading" licenses to the latest version release at special client rates.

Recommended minimum hardware configuration for the **DYNATUNE-XL** Tools are Intel Core i5/i7 CPU (or similar) with 4GB Ram.

All Units in **DYNATUNE-XL** are SI.

DYNATUNE-XL DEMO VERSIONS

DEMO Versions of the following DYNATUNE-XL Modules can be downloaded here:

- DYNATUNE Ride &Handling Module: http://www.dynatune-xl.com/download-demo-rh.html
- DYNATUNE Suspension Design Module: http://www.dynatune-xl.com/download-demo-sdm.html
- DYNATUNE Suspension Tuning Module: http://www.dynatune-xl.com/download-demo-stm.html

DYNATUNE-XL STORE

B2C customers can acquire the various **DYNATUNE-XL** Modules online in our webstore:

http://www.dynatune-xl.com/store/c1/Featured_Products.html

B2B customers are kindly requested to contact us directly.

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